

Problem

Fix

Goal

Overfitting
memorises the
training data

Early stopping
Note ML-057, DL-022

Dropout
Note DL-024–025

L1 and L2
regularisation
Note DL-026

Unstable gradients
vanishing or exploding
Note DL-018

Better activations
Note DL-027

ReLU and
its variants
Note DL-028

Weight initialisation
Note DL-029

Xavier and He
Note DL-030

Batch normalisation
Note DL-031

Backpropagation
Note DL-015–017

Slow training
poor or wobbly
convergence

Scale the inputs
Note DL-023

Choose the batch size
Note DL-020

Better optimizers
see the optimizers map

**A better
neural network**
Note DL-021

→ fixes

→ is a kind of / leads to

→ makes the network better

from another map